

peak performance at around 125 characters. These results indicate that the probes remain reliable even for relatively short texts.

C.8 MGT Probes performance on AI-edit text detection

Model	Accuracy	Macro F1	Human F1	AI-edited F1	AI-generated F1
RoBERTa	0.652	0.642	0.583	0.473	0.869
EditLens	0.525	0.438	0.575	0.739	0.000
LLP	0.848	0.848	0.806	0.754	0.985
CLP	0.854	0.853	0.811	0.764	0.985

Table 5: Three-way classification on EditLens (Thai et al., 2026), comparing the EditLens Llama-3.2-3B model, RoBERTa (Liu et al., 2019), and both MGT probes. The three classes are “AI-generated”, “AI-edited”, and “Human”.

Figure 6 shows a positive correlation between edit strength and the probe projection scores. In Table 5, we present initial results on three-way AI-edit classification using EditLens (Thai et al., 2026). We compare two baselines, RoBERTa (Liu et al., 2019) and the EditLens Llama-3.2-3B model,²² against our MGT probes.

We split the EditLens data into train/validation/test sets of 1,000/200/200 examples. We train the probes and RoBERTa on the training set and calibrate all three methods on the validation set by selecting two thresholds for the classes “AI-generated”, “AI-edited”, and “Human”. We do not further train the EditLens model, as it was already trained on this dataset.

Table 5 reports accuracy and class-wise F1 scores. Both probes achieve strong classification performance relative to the baselines, despite not being explicitly trained for three-way classification. While these results indicate that the probe scores may support fine-grained AI-edit classification, the analysis is preliminary and does not support broader conclusions. We leave a more comprehensive evaluation of this direction to future work.

C.9 Calibration Analysis

Table 6 presents a small-scale calibration analysis of the MGT probes, comparing their Expected Calibration Error (ECE) against RoBERTa (Liu et al., 2019). The results show that both probes are well calibrated across DetectRL (Wu et al., 2024) subsets and, in several settings, outperform a fully fine-tuned RoBERTa model in calibration.

²²https://huggingface.co/pangram/editlens_Llama-3.2-3B

Model	ArXiv	Reddit	Yelp	News
LLP	0.005	0.009	0.010	0.007
CLP	0.000	0.006	0.007	0.003
RoBERTa	0.002	0.006	0.033	0.002

Table 6: Expected Calibration Error (ECE) analysis comparing MGT probes against RoBERTa on DetectRL (Wu et al., 2024).

We emphasize that this analysis is performed on the projection scores produced by the linear probes. These results provide additional evidence that the probe outputs are well calibrated and support their potential use for fine-grained AI-edit detection.

D Ablation Studies

Lastly, we conduct a series of ablation experiments for both probing variants LLP and CLP.

D.1 LLP

D.1.1 Model Size and Architecture Ablations

Setting	TSM (Quaremba et al., 2026)			
	FP	PE	SUM	TST
Baseline	0.971	0.942	0.978	0.896
Token Aggregation				
Pooling	-0.019	-0.155	-0.021	-0.085
Layer Selection				
First layer	-0.388	-0.316	-0.431	-0.355
Last layer	-0.036	-0.077	-0.038	-0.065
Model				
Llama-3B	+0.001	-0.012	+0.006	+0.000
Llama-1B	-0.011	-0.034	+0.004	-0.011
Qwen-32B	+0.020	+0.029	+0.013	+0.048
Qwen-8B	+0.016	+0.016	+0.010	+0.031
Qwen-4B	+0.015	+0.011	+0.007	+0.029
Qwen-0.6B	-0.012	-0.036	-0.005	-0.022
Regularization Penalty				
C=0.01	-0.001	+0.001	+0.000	+0.001
C=0.1	+0.000	+0.001	+0.000	+0.002
C=10	+0.001	+0.001	+0.000	+0.002
PCA Activations				
k=10	-0.064	-0.117	-0.046	-0.120
k=50	-0.009	-0.015	-0.008	-0.022
k=150	+0.004	+0.005	+0.001	+0.011
k=200	+0.003	+0.006	-0.001	+0.011
k=250	+0.003	+0.006	-0.002	+0.010
No PCA	+0.016	+0.010	+0.007	+0.014

Table 7: Ablation study of LLP across five design choices.

Table 7 reports AUC scores for the LLP variant defined in Section 4, covering five design choices: (1) token aggregation (last token vs. mean pooling), (2) layer selection (first layer, last layer, or layer-wise aggregation), (3) model architecture and size,

(4) regularization strength, and (5) the number of PCA components. We restrict this analysis to TSM, as it consistently proved to be the most challenging benchmark throughout our experiments.

First, probing only the final token yields moderate but consistent improvements over mean pooling. This likely reflects the fact that the final token aggregates information from the entire sequence through causal attention.

Second, using only the first or last layer instead of aggregating projection scores across layers consistently reduces performance. This finding aligns with Nordby et al. (2026), who show that ensembles of probes outperform single-layer probes. We do not include a single middle-layer probe, as TextFluoroscopy (Yu et al., 2024) already optimizes for the best-performing layer, and our results demonstrate that LLP consistently outperforms this approach.

Third, we evaluate the impact of model architecture and size by replacing LLaMA with Qwen (Yang et al., 2025). We find that Qwen-8B consistently improves performance over LLaMA-8B. Notably, even Qwen-4B achieves slightly stronger results, despite its smaller size. These findings suggest that architecture may play a more important role than parameter count and motivate future work on how model design influences the linear separability of MGT and HWT. Within the LLaMA family, reducing model size leads to only modest performance degradation.

Fourth, varying the strength of the L_2 regularization penalty has virtually no effect on performance, suggesting that the learned MGT direction is stable and readily recoverable.

Finally, motivated by our finding that MGT and HWT are linearly separable in low-dimensional subspaces (§3), we vary the number of principal components used to project activations before training probes. Increasing the dimensionality beyond $k = 100$ yields only marginal and inconsistent improvements. This result further supports our representation analysis, indicating that MGT signals are encoded in relatively low-dimensional regions of the activation space.

D.1.2 Detailed PCA vs no-PCA comparison

Mode	Multisocial				DetectRL Domains				RAID Models			
	en	de	ru	zh	ArXiv	Reddit	Yelp	News	Cohere	GPT-4	Llama	Mistral
LLP (No PCA)	0.963	0.933	0.915	0.950	1.000	1.000	1.000	1.000	0.969	0.992	0.995	0.989
LLP (PCA)	0.957	0.934	0.921	0.944	1.000	1.000	1.000	1.000	0.971	0.992	0.995	0.993
LLP (No PCA) - LLP (PCA)	0.006	-0.001	-0.006	0.006	0.000	0.000	0.000	0.000	-0.002	-0.001	0.001	-0.004

Table 8: LLP ablation study comparing PCA and no-PCA variants. The PCA variant is the main specification, using 100 dimensions.

A potential concern is that PCA may artificially facilitate linear separability, for example by removing noisy dimensions. While Table 7 already shows nearly identical probe performance with and without PCA, we provide a more detailed comparison in Table 8. Specifically, we compare the no-PCA variant against the PCA variant using the top 100 principal components, as in our main experiments. The near-identical results indicate that PCA does not meaningfully affect linear separability. Rather, it primarily serves as a dimensionality-reduction step that reduces the computational and memory requirements of the probes.

D.2 CLP

D.2.1 Model Size and Architecture Ablations

Setting	TSM (Quaremba et al., 2026)			
	FP	PE	SUM	TST
Baseline	0.973	0.920	0.970	0.890
Token Aggregation				
Pooling	-0.008	-0.088	+0.001	-0.036
Model				
Llama-3B	-0.001	-0.019	-0.002	-0.021
Llama-1B	-0.016	-0.033	-0.002	-0.015
Qwen-32B	+0.016	+0.029	+0.012	+0.025
Qwen-8B	+0.012	+0.001	+0.010	+0.020
Qwen-4B	+0.005	+0.018	+0.010	+0.009
Qwen-0.6B	-0.012	-0.016	-0.003	-0.043
Regularization Penalty				
C=0.01	+0.001	+0.016	+0.005	+0.003
C=0.1	+0.001	+0.009	+0.003	+0.000
C=10	-0.001	-0.001	-0.000	-0.011
PCA Activations				
k=10	-0.036	-0.010	-0.009	-0.056
k=50	-0.004	+0.000	-0.010	-0.014
k=150	-0.002	+0.003	-0.000	+0.002
k=200	+0.001	+0.004	+0.001	+0.003
k=250	+0.003	+0.004	+0.002	+0.003
No PCA	+0.015	+0.031	+0.012	+0.026

Table 9: Ablation study of CLP across four design choices.

Table 9 presents CLP ablations across four design choices: (1) token aggregation (last token vs. mean pooling), (2) model architecture and size, (3) regularization strength, and (4) the number of PCA

components.

Overall, we observe trends similar to those for LLP in Table 7. Using the last token consistently outperforms mean pooling. Smaller Llama variants reduce performance, whereas Qwen yields modest but consistent improvements. The regularization penalty has only a minor effect at $\lambda = 0.001$ and a negligible impact for other values. Finally, as with LLP, 100 PCA components provide the best trade-off between performance and dimensionality, with no meaningful gains from using additional components.

D.2.2 Detailed PCA vs no-PCA comparison

Mode	Multisocial				DetectRL Domains				RAID Models			
	en	de	ru	zh	ArXiv	Reddit	Yelp	News	Cohere	GPT-4	Llama	Mistral
CLP (PCA)	0.951	0.920	0.906	0.951	1.000	1.000	1.000	1.000	0.965	0.991	0.995	0.987
CLP (No PCA)	0.962	0.937	0.919	0.951	1.000	1.000	1.000	1.000	0.967	0.989	0.995	0.987
CLP (PCA) - CLP (No PCA)	-0.011	-0.017	-0.013	0.000	0.000	0.000	-0.000	0.000	-0.002	0.002	-0.000	0.000

Table 10: CLP ablation study comparing PCA and no-PCA variants. The PCA variant is the main specification, using 100 dimensions.

Table 10 reports the same PCA versus no-PCA comparison for CLP. Consistent with the LLP results, performance is nearly identical across both variants, indicating that PCA does not meaningfully affect linear separability for either probe.